Name: K R Lavanya

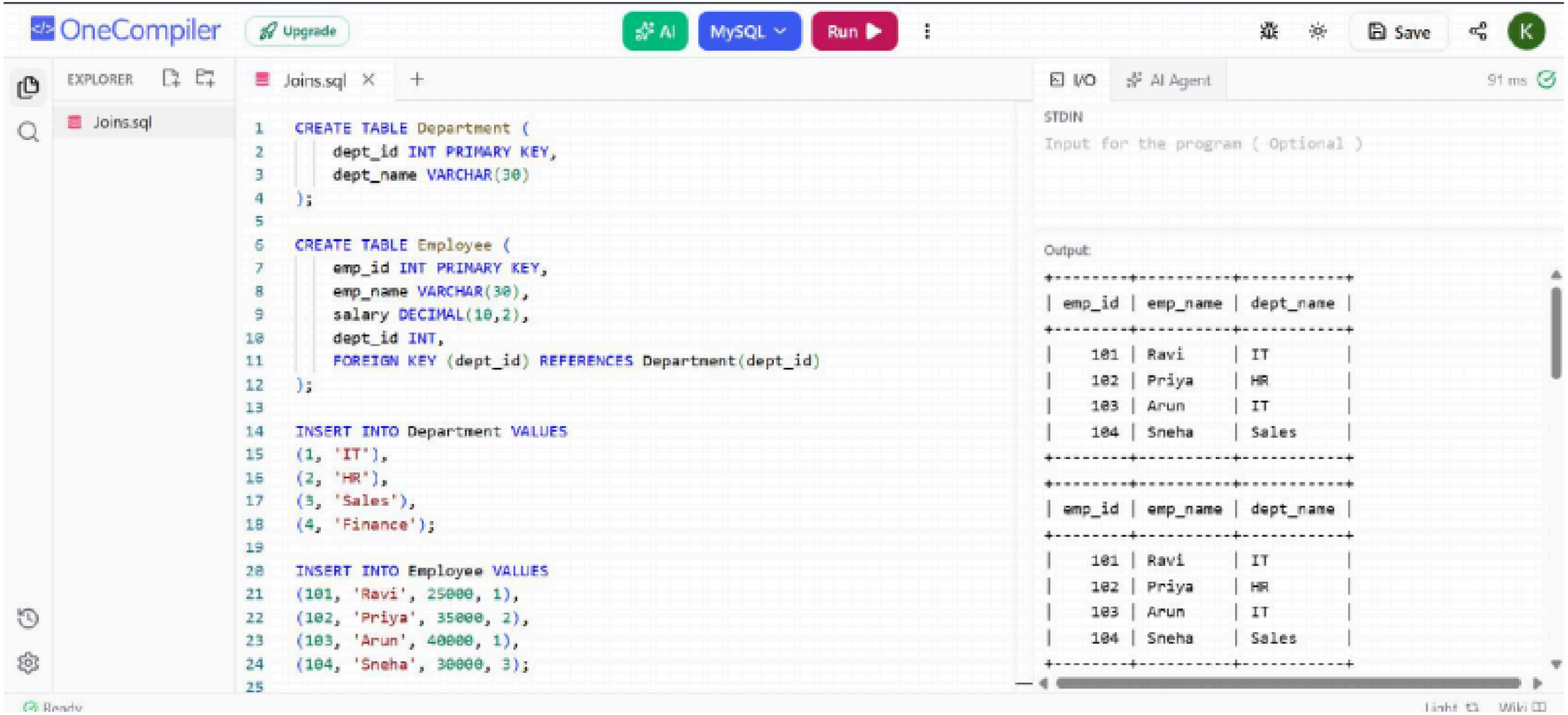
USN:1RUA25BCA0045

Lab 7

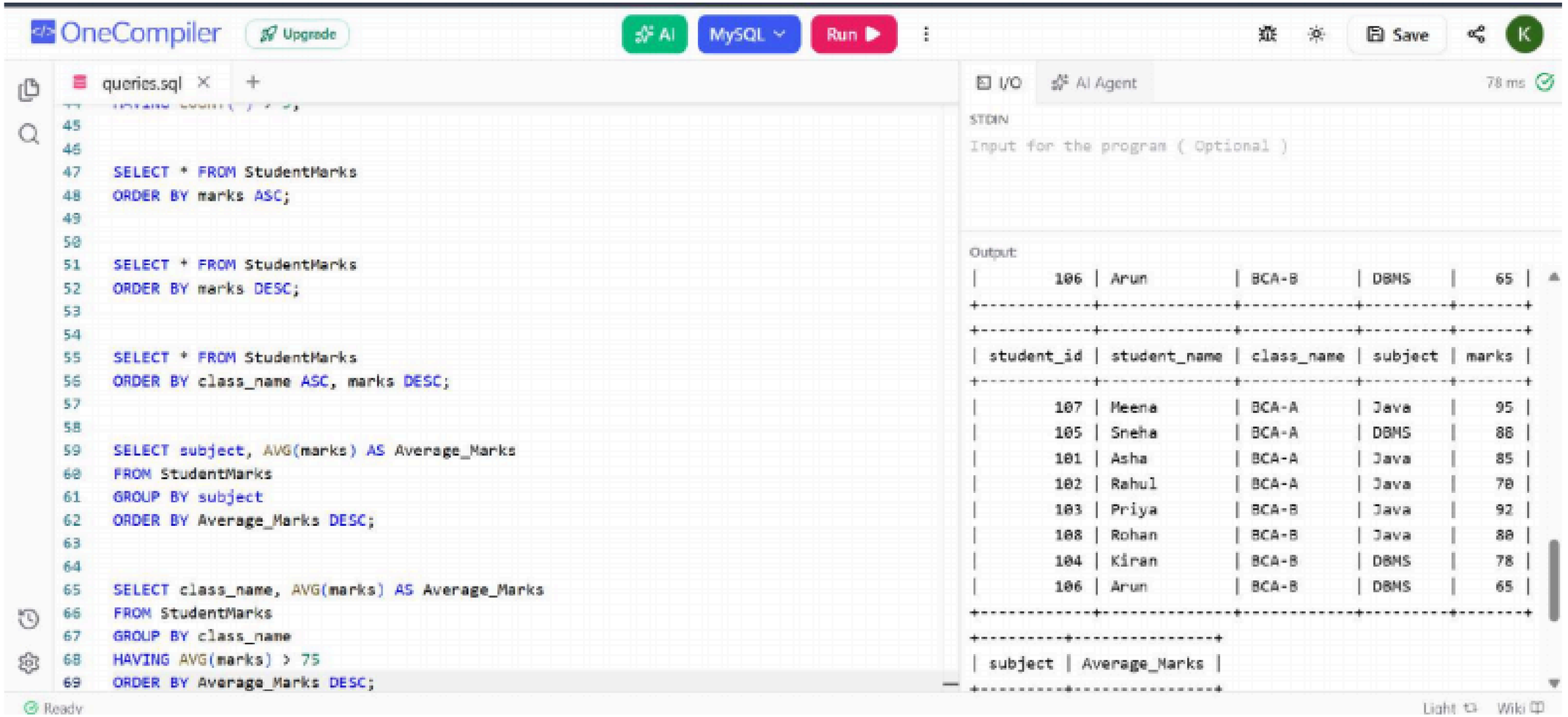



```
43
44
45  SELECT Employee.emp_id, Employee.emp_name,
46         Department.dept_name
47  FROM Employee
48  LEFT JOIN Department
49  ON Employee.dept_id = Department.dept_id
50
51  UNION
52
53  SELECT Employee.emp_id, Employee.emp_name,
54         Department.dept_name
55  FROM Employee
56  RIGHT JOIN Department
57  ON Employee.dept_id = Department.dept_id;
58
59
60  SELECT emp_id, emp_name, salary, dept_id, dept_name
61  FROM Employee
62  NATURAL JOIN Department;
63
64
65  SELECT Employee.emp_name, Department.dept_name
66  FROM Employee
67  CROSS JOIN Department;
```

```
| emp_id | emp_name | dept_name |
+--------+----------+-----------+
|    101 | Ravi     | IT        |
|    102 | Priya    | HR        |
|    103 | Arun     | IT        |
|    104 | Sneha    | Sales     |
|   NULL | NULL     | Finance   |
+--------+----------+-----------+
+--------+----------+----------+---------+-----------+
| emp_id | emp_name | salary   | dept_id | dept_name |
+--------+----------+----------+---------+-----------+
|    101 | Ravi     | 25000.00 |       1 | IT        |
|    102 | Priya    | 35000.00 |       2 | HR        |
|    103 | Arun     | 40000.00 |       1 | IT        |
|    104 | Sneha    | 30000.00 |       3 | Sales     |
+--------+----------+----------+---------+-----------+
+----------+-----------+
```

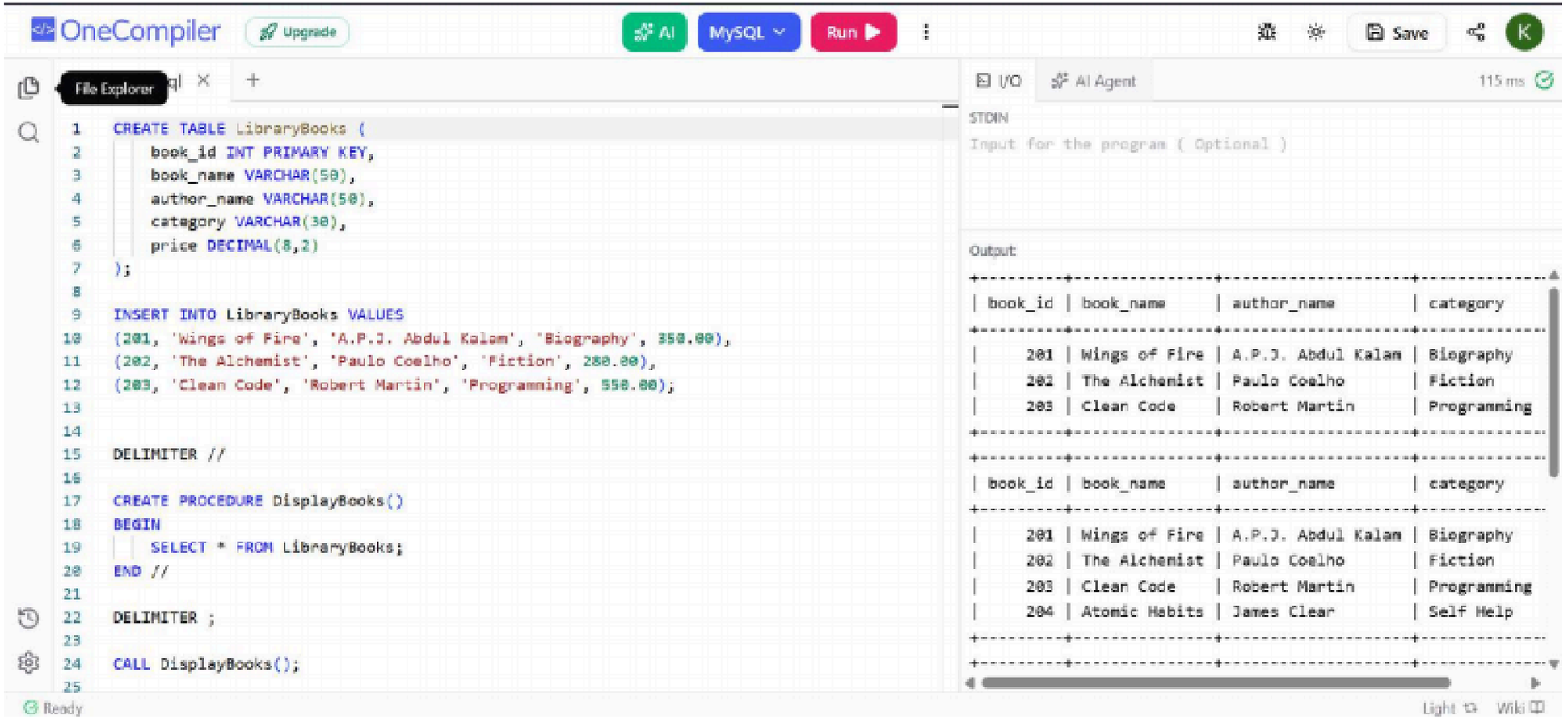

OneCompiler
Upgrade
AI
MySQL
Run
Save
EXPLORER
Joins.sql
Joins.sql
I/O
AI Agent
91 ms
STDIN
Input for the program (Optional)
Output:
CREATE TABLE Department (
dept_id INT PRIMARY KEY,
dept_name VARCHAR(30)
);
CREATE TABLE Employee (
emp_id INT PRIMARY KEY,
emp_name VARCHAR(30),
salary DECIMAL(10,2),
dept_id INT,
FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);
INSERT INTO Department VALUES
(1, 'IT'),
(2, 'HR'),
(3, 'Sales'),
(4, 'Finance');
INSERT INTO Employee VALUES
(101, 'Ravi', 25000, 1),
(102, 'Priya', 35000, 2),
(103, 'Arun', 40000, 1),
(104, 'Sneha', 30000, 3);
+--------+----------+-----------+
| emp_id | emp_name | dept_name |
+--------+----------+-----------+
101	Ravi	IT
102	Priya	HR
103	Arun	IT
104	Sneha	Sales
+--------+----------+-----------+
+--------+----------+-----------+
| emp_id | emp_name | dept_name |
+--------+----------+-----------+
101	Ravi	IT
102	Priya	HR
103	Arun	IT
104	Sneha	Sales
+--------+----------+-----------+

Lab 8


OneCompiler
Upgrade
AI
MySQL
Run
Save
queries.sql
I/O
AI Agent
78 ms
STDIN
Input for the program (Optional)
Output:
SELECT * FROM StudentMarks
ORDER BY marks ASC;
SELECT * FROM StudentMarks
ORDER BY marks DESC;
SELECT * FROM StudentMarks
ORDER BY class_name ASC, marks DESC;
SELECT subject, AVG(marks) AS Average_Marks
FROM StudentMarks
GROUP BY subject
ORDER BY Average_Marks DESC;
SELECT class_name, AVG(marks) AS Average_Marks
FROM StudentMarks
GROUP BY class_name
HAVING AVG(marks) > 75
ORDER BY Average_Marks DESC;
106	Arun	BCA-B	DBMS	65
student_id	student_name	class_name	subject	marks
107	Meena	BCA-A	Java	95
105	Sneha	BCA-A	DBMS	88
101	Asha	BCA-A	Java	85
102	Rahul	BCA-A	Java	70
103	Priya	BCA-B	Java	92
108	Rohan	BCA-B	Java	80
104	Kiran	BCA-B	DBMS	78
106	Arun	BCA-B	DBMS	65
subject	Average_Marks
Ready
Light
Wiki


OneCompiler
Upgrade
AI
MySQL
Run
Save
queries.sql
I/O
AI Agent
78 ms
STDIN
Input for the program (Optional)
Output:
CREATE TABLE StudentMarks (
student_id INT PRIMARY KEY,
student_name VARCHAR(30),
class_name VARCHAR(20),
subject VARCHAR(30),
marks INT
);
INSERT INTO StudentMarks VALUES
(101, 'Asha', 'BCA-A', 'Java', 85),
(102, 'Rahul', 'BCA-A', 'Java', 70),
(103, 'Priya', 'BCA-B', 'Java', 92),
(104, 'Kiran', 'BCA-B', 'DBMS', 78),
(105, 'Sneha', 'BCA-A', 'DBMS', 88),
(106, 'Arun', 'BCA-B', 'DBMS', 65),
(107, 'Meena', 'BCA-A', 'Java', 95),
(108, 'Rohan', 'BCA-B', 'Java', 80);
SELECT subject, AVG(marks) AS Average_Marks
FROM StudentMarks
GROUP BY subject;
SELECT class_name, COUNT(*) AS Total_Students
+---------+---------------+
| subject | Average_Marks |
+---------+---------------+
| Java | 84.4000 |
| DBMS | 77.0000 |
+---------+---------------+
+------------+----------------+
| class_name | Total_Students |
+------------+----------------+
| BCA-A | 4 |
| BCA-B | 4 |
+------------+----------------+
+---------+-------------+
| subject | Total_Marks |
+---------+-------------+
| Java | 422 |
Ready
Light
Wiki

Lab 9




```
CREATE TABLE LibraryBooks (
    book_id INT PRIMARY KEY,
    book_name VARCHAR(50),
    author_name VARCHAR(50),
    category VARCHAR(30),
    price DECIMAL(8,2)
);

INSERT INTO LibraryBooks VALUES
(201, 'Wings of Fire', 'A.P.J. Abdul Kalam', 'Biography', 350.00),
(202, 'The Alchemist', 'Paulo Coelho', 'Fiction', 280.00),
(203, 'Clean Code', 'Robert Martin', 'Programming', 550.00);

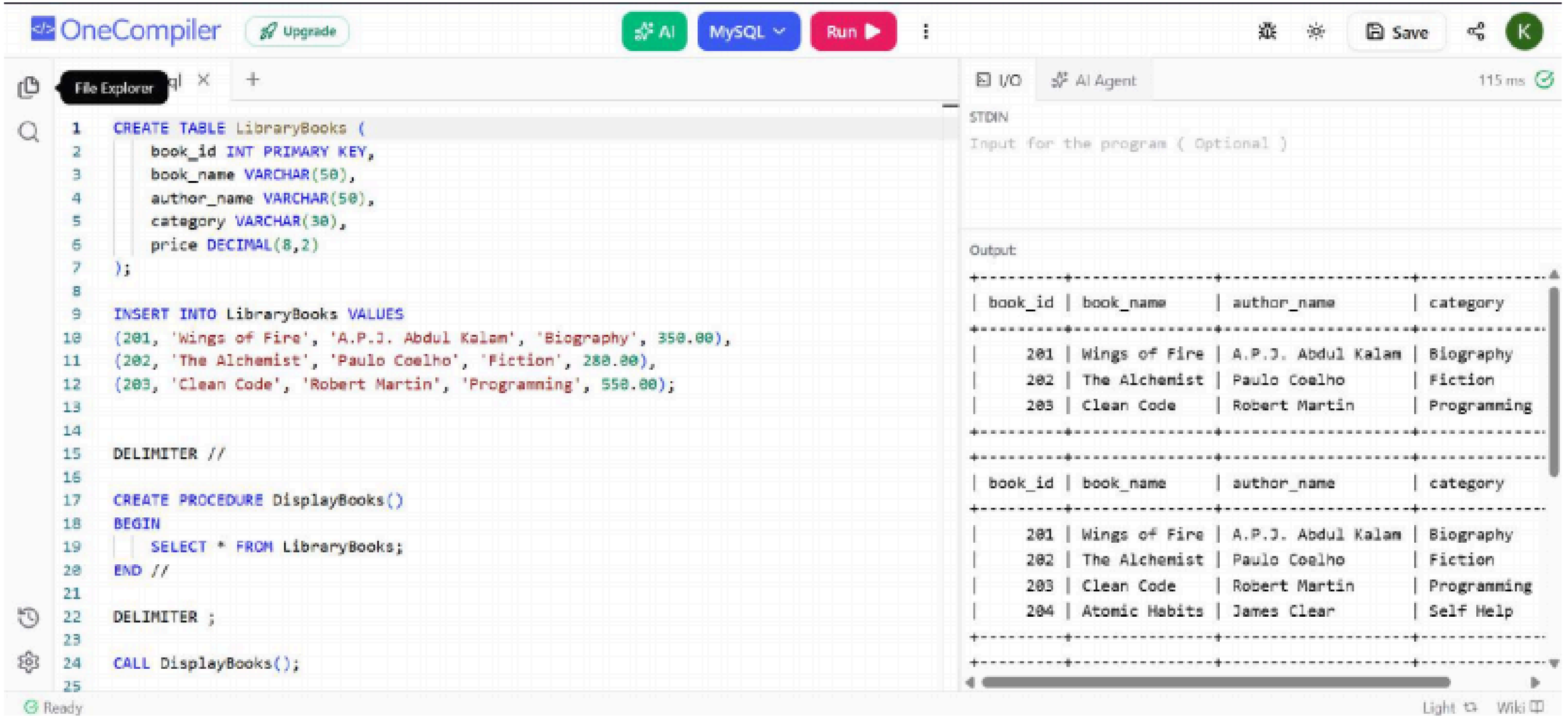

DELIMITER //

CREATE PROCEDURE DisplayBooks()
BEGIN
    SELECT * FROM LibraryBooks;
END //

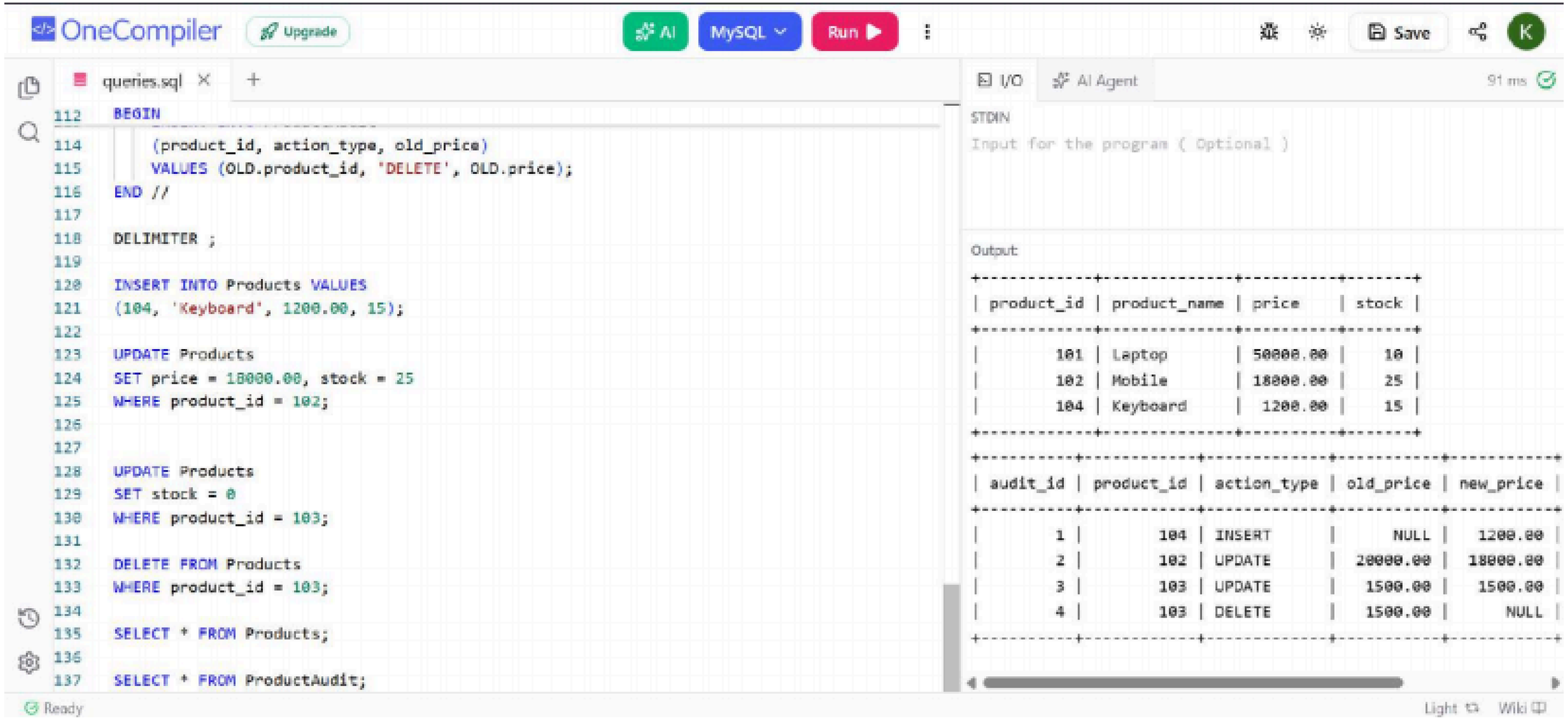
DELIMITER ;

CALL DisplayBooks();
```

```
+---------+--------------+--------------------+-------------+
| book_id | book_name    | author_name        | category    |
+---------+--------------+--------------------+-------------+
|     201 | Wings of Fire | A.P.J. Abdul Kalam | Biography   |
|     202 | The Alchemist | Paulo Coelho       | Fiction     |
|     203 | Clean Code    | Robert Martin      | Programming |
+---------+--------------+--------------------+-------------+
+---------+--------------+--------------------+-------------+
| book_id | book_name    | author_name        | category    |
+---------+--------------+--------------------+-------------+
|     201 | Wings of Fire | A.P.J. Abdul Kalam | Biography   |
|     202 | The Alchemist | Paulo Coelho       | Fiction     |
|     203 | Clean Code    | Robert Martin      | Programming |
|     204 | Atomic Habits | James Clear        | Self Help   |
+---------+--------------+--------------------+-------------+
```


OneCompiler
Upgrade
AI
MySQL
Run
Save
File Explorer
I/O
AI Agent
115 ms
STDIN
Input for the program (Optional)
Output
CREATE TABLE LibraryBooks (
book_id INT PRIMARY KEY,
book_name VARCHAR(50),
author_name VARCHAR(50),
category VARCHAR(30),
price DECIMAL(8,2)
);
INSERT INTO LibraryBooks VALUES
(201, 'Wings of Fire', 'A.P.J. Abdul Kalam', 'Biography', 350.00),
(202, 'The Alchemist', 'Paulo Coelho', 'Fiction', 280.00),
(203, 'Clean Code', 'Robert Martin', 'Programming', 550.00);
DELIMITER //
CREATE PROCEDURE DisplayBooks()
BEGIN
SELECT * FROM LibraryBooks;
END //
DELIMITER ;
CALL DisplayBooks();
+---------+--------------+--------------------+-------------+
| book_id | book_name | author_name | category |
+---------+--------------+--------------------+-------------+
201	Wings of Fire	A.P.J. Abdul Kalam	Biography
202	The Alchemist	Paulo Coelho	Fiction
203	Clean Code	Robert Martin	Programming
+---------+--------------+--------------------+-------------+
+---------+--------------+--------------------+-------------+
| book_id | book_name | author_name | category |
+---------+--------------+--------------------+-------------+
201	Wings of Fire	A.P.J. Abdul Kalam	Biography
202	The Alchemist	Paulo Coelho	Fiction
203	Clean Code	Robert Martin	Programming
204	Atomic Habits	James Clear	Self Help
+---------+--------------+--------------------+-------------+
Ready
Light
Wiki

Lab 10




```
BEGIN
    (product_id, action_type, old_price)
    VALUES (OLD.product_id, 'DELETE', OLD.price);
END //

DELIMITER ;

INSERT INTO Products VALUES
(104, 'Keyboard', 1200.00, 15);

UPDATE Products
SET price = 18000.00, stock = 25
WHERE product_id = 102;

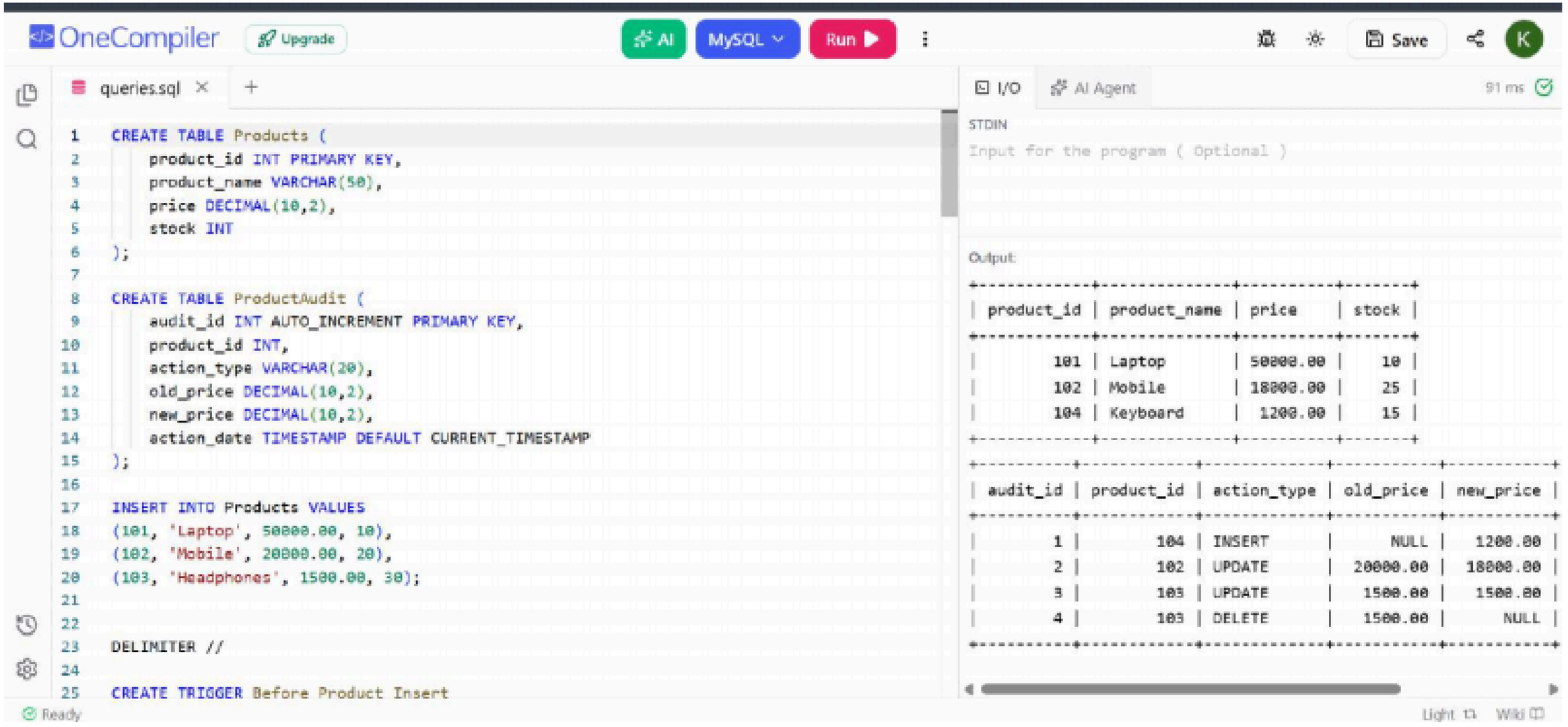

UPDATE Products
SET stock = 0
WHERE product_id = 103;

DELETE FROM Products
WHERE product_id = 103;

SELECT * FROM Products;

SELECT * FROM ProductAudit;
```

```
+------------+--------------+----------+-------+
| product_id | product_name | price    | stock |
+------------+--------------+----------+-------+
|        101 | Laptop       | 50000.00 |    10 |
|        102 | Mobile       | 18000.00 |    25 |
|        104 | Keyboard     |  1200.00 |    15 |
+------------+--------------+----------+-------+
+----------+------------+-------------+-----------+-----------+
| audit_id | product_id | action_type | old_price | new_price |
+----------+------------+-------------+-----------+-----------+
|        1 |        104 | INSERT      |      NULL |   1200.00 |
|        2 |        102 | UPDATE      |  20000.00 |  18000.00 |
|        3 |        103 | UPDATE      |   1500.00 |   1500.00 |
|        4 |        103 | DELETE      |   1500.00 |      NULL |
+----------+------------+-------------+-----------+-----------+
```




```sql
CREATE TABLE Products (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(50),
    price DECIMAL(10,2),
    stock INT
);

CREATE TABLE ProductAudit (
    audit_id INT AUTO_INCREMENT PRIMARY KEY,
    product_id INT,
    action_type VARCHAR(20),
    old_price DECIMAL(10,2),
    new_price DECIMAL(10,2),
    action_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

INSERT INTO Products VALUES
(101, 'Laptop', 50000.00, 10),
(102, 'Mobile', 20000.00, 20),
(103, 'Headphones', 1500.00, 30);


DELIMITER //

CREATE TRIGGER Before Product Insert
```

```
+------------+--------------+----------+-------+
| product_id | product_name | price    | stock |
+------------+--------------+----------+-------+
|        101 | Laptop       | 50000.00 |    10 |
|        102 | Mobile       | 18000.00 |    25 |
|        104 | Keyboard     |  1200.00 |    15 |
+------------+--------------+----------+-------+

+----------+------------+-------------+-----------+-----------+
| audit_id | product_id | action_type | old_price | new_price |
+----------+------------+-------------+-----------+-----------+
|        1 |        104 | INSERT      |      NULL |   1200.00 |
|        2 |        102 | UPDATE      |  20000.00 |  18000.00 |
|        3 |        103 | UPDATE      |   1500.00 |   1500.00 |
|        4 |        103 | DELETE      |   1500.00 |      NULL |
+----------+------------+-------------+-----------+-----------+
```